Simulation-Based Analysis of Shear Stress in Granular Materials using LAMMPS

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Project Repository: `https://github.com/Siddhantkumar/Granular-Shear-Analysis`

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1 Introduction

Dense granular materials exhibit a complex transition between solid-like and fluid-like behaviour under shear. In engineering and geomechanics, the stress response of a granular assembly determines stability in processes such as hopper flow, powder compaction, and shear-driven failure. The present project uses a simulation-driven workflow to generate physically consistent packings, compress them to a dense state, and then impose shear at controlled rates to analyze the resulting stress, velocity, and concentration fields.

The workflow combines discrete element modeling in LAMMPS with MATLAB-based coarse-graining and visualization. This approach permits direct interpretation of per-particle dynamics through continuum-level fields, including normal and shear stress components, local shear rate, concentration gradients, and pressure distribution.

2 Problem Statement

The objective of this study is to characterize the shear stress behaviour of a dense granular assembly as a function of shear rate. The repository contains the sequential stages necessary to produce dense packings, shear them at different rates, and extract mechanical fields. The central problem is to understand how the granular stress tensor and associated flow fields evolve across the shear rate cases.

This study is important because granular flow rheology is rate-dependent and exhibits nontrivial interactions between contact forces, shear deformation, and particle concentration. Accurate modeling of these effects is essential for predicting failure and designing granular handling systems.

3 Objective

The project aims to:

- Generate an initial granular packing by pouring spherical particles under gravity.
- Compress the poured configuration to a target pressure using a wall-feedback algorithm.
- Perform shear simulations at multiple shear rates and obtain per-particle kinematic and stress data.
- Post-process the resulting data using MATLAB to compute bulk stresses, velocity gradients, concentration fields, and pressure profiles.
- Compare the results across the available shear rate cases and interpret trends in stress response.

4 Methodology

The simulation workflow is composed of four stages: pouring, compression, shearing, and MATLAB post-processing. Each stage is implemented through specific repository files.

4.1 Pouring Simulation

The initial granular packing is created by the LAMMPS pouring code. Key simulation choices are:

- **Units and geometry:** SI units, two-dimensional simulation, sphere atom style.
- **Boundary conditions:** periodic in the z direction, fixed in y, and shrink-wrap in x to allow box deformation during packing.
- **Particle generation:** particle diameter $d = 0.002$ m, density 2500 kg/m^3, created within a drop region defined by the box dimensions 0.1×0.032 m.
- **Contact model:** granular Hooke contact with history dependence, using stiffness parameters $k_n = 10^5$ N/m, $k_t = 2 \times 10^4$ N/m, normal damping $\gamma_n = 50$ and tangential damping $\gamma_t = 15$.
- **Dynamics:** fixed-mass integration with `fix integrate all nve/sphere`, gravity $9.81\,\mathrm{m/s^2}$ in the negative y direction, and two-dimensional enforcement via `fix z_enforce`.

The final packed configuration is saved as a restart file `pour file/restart3.poured` after 50,000 timesteps. This file represents the initial particle arrangement for subsequent compression.

4.2 Compression Stage

The next stage is performed by the LAMMPS restart file that we got from previous pouring stage. This script reads the poured data file and compresses the assembly using a pressure-controlled top wall. The main elements are:

- **Read input:** the script reads `Original_data.pour` and resizes the box in y to create a free gap above the grains.
- **Contact mechanics:** a lower stiffness $k_n = 400$ N/m and vanishing tangential stiffness $k_t = 0$ are used for wall contacts, with damping scaled to the particle diameter.
- **Pressure computation:** per-atom stress is computed with `compute peratom all stress/atom NULL`. The instantaneous normal pressure on the top wall is obtained as

$$P_{inst} = -\frac{\sum_i c_peratom[2]_i}{l_x l_y} \tag{1}$$

  where $c_peratom[2]$ is the yy component of the atomic stress and $l_x l_y$ is the area in the simulation plane.
- **Feedback loop:** a proportional controller adjusts the top wall position based on the pressure error $P_{target} - P_{inst}$, with a maximum wall displacement cap of 5×10^{-6} m per cycle. The wall is moved with "fix deform 1 y delta 0.0 vymove remap none" to compress the sample while preserving particle positions.

The compression loop repeats for 10,000 iterations, and the final compressed configuration is saved as `compressed_grains.data`. This file is the input for all subsequent shear simulations.

4.3 Shear Simulation Stage

The repository contains shear simulation results in folders named `sr_0.1`, `sr_0.2`, and `sr_0.5`. Each folder corresponds to a different imposed shear rate, $\dot{\gamma} = 0.1$, 0.2, and 0.5 s^{-1} respectively.

Within each shear-rate folder, the following files document the state after shear equilibration:

- `dump_after_eqm.shear` – a shear-phase dump file likely containing the particle configuration after equilibrium.
- `raw_pressure.dat` – a time series of instantaneous pressure values measured during the shear simulation.
- `sheared_compressed_grains.data` – a LAMMPS data file containing the sheared particle configuration and properties.

A sample of `sr_0.1/raw_pressure.dat` shows a header and two columns, with the second column representing the instantaneous normal pressure v_P_inst recorded at each timestep. The pressure values settle near a stable value, indicating that the shear test maintained a controlled normal load.

The sheared data file `sr_0.1/sheared_compressed_grains.data` has the standard LAMMPS dump structure with 5000 atoms, three atom types, box bounds, and particle attributes such as position, diameter, and mass. This confirms the use of discrete element output as the base for post-processing.

4.4 Data Extraction and Post-processing

The MATLAB script contains the post-processing pipeline used across the shear-rate cases. It reads the LAMMPS atom dump files that we get from the LAMMPS simulation for shearing with pressure feedback, and computes spatially averaged fields along the vertical direction.

The raw output files follow the format: `ITEM: ATOMS id type x y z vx vy vz diameter mass c_peratom[1] c_peratom[2] c_peratom[4]` These files provide particle positions, velocity components, diameter, mass, and per-atom stress components. In the MATLAB processing script, the bulk stress components are computed as:

$$S_{xx}(t) = \frac{1}{A}\sum_i c_peratom[1]_i, \tag{2}$$

$$S_{yy}(t) = \frac{1}{A}\sum_i c_peratom[2]_i, \tag{3}$$

$$S_{xy}(t) = \frac{1}{A}\sum_i c_peratom[4]_i, \tag{4}$$

where $A = L_x L_y$ is the instantaneous cross-sectional area of the sample. The MATLAB script then stores the time series $(t, S_{xx}, S_{yy}, S_{xy})$ in `stress_history_output.txt`.

The script also computes coarse-grained vertical profiles using a Lucy weighting kernel. For each vertical slice, it evaluates the effective cross-sectional contribution of grains and aggregates:

- V_x – local average streamwise velocity,
- ∇V_x – local shear rate,
- $\nabla^2 V_x$ – velocity curvature,
- P – local pressure (mean normal stress),
- ∇P – pressure gradient.

The script uses a coarse-graining window width $W = 4d$ and a slice count of $N_{slice} = 600$. The spatial coordinate is normalized by the particle diameter through the vector `z_norm`. This allows the generation of continuum profiles from the discrete particle data.

4.5 Visualization

The MATLAB processing produces three figures for each shear rate case:

1. Kinematics and concentration profiles, including $V_x(z)$, $\dot{\gamma}(z)$.
2. Forces and scale fields, including $\nabla^2 V_x(z)$, the pressure profile $P(z)$, and the pressure gradient.
3. Stress evolution versus time, showing $S_{xx}(t)$, $S_{yy}(t)$, $S_{xy}(t)$, and the effective friction coefficient S_{xy}/S_{yy}.

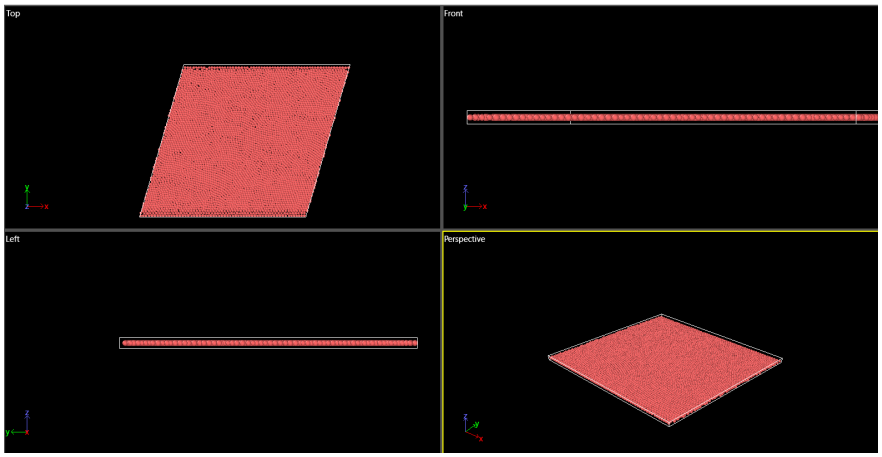
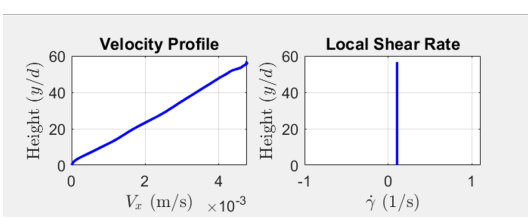


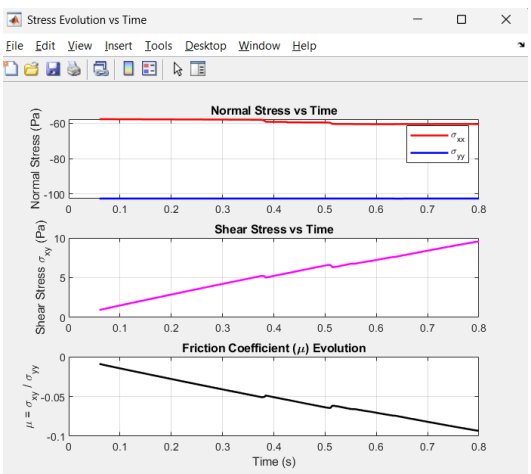
Figure 1: Visualisation in Ovito

5 Results

These images correspond to the MATLAB post-processing outputs for shear rates 0.1, 0.2, and 0.5 s^{-1}.

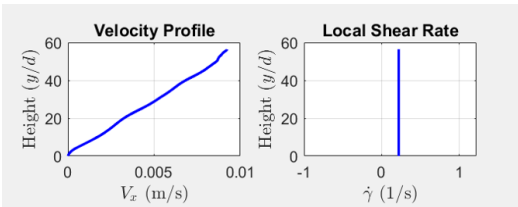


(a) Velocity profile for $\dot{\gamma} = 0.1$.

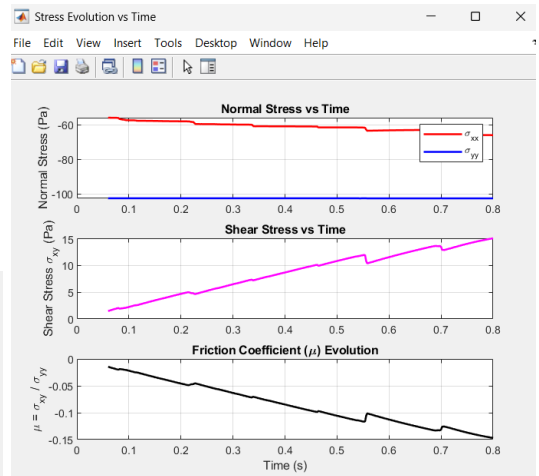


(b) Stress evolution for $\dot{\gamma} = 0.1$.

Figure 2: MATLAB post-processing results for Shear rate 0.1 s^{-1}.

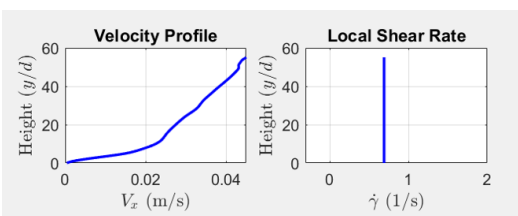


(a) Velocity profile for $\dot{\gamma} = 0.2$.

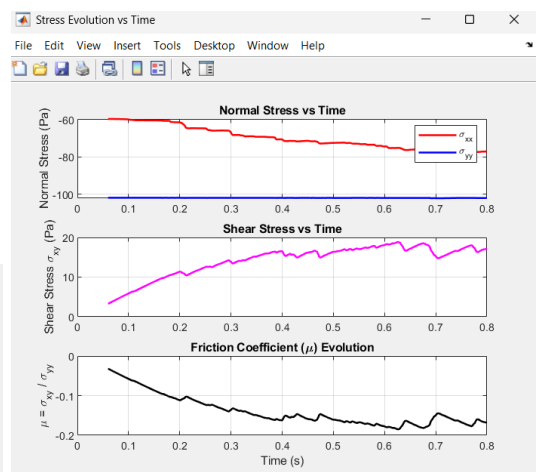


(b) Stress evolution for $\dot{\gamma} = 0.2$.

Figure 3: MATLAB post-processing results for Shear rate 0.2 s^{-1}



(a) Velocity profile for $\dot{\gamma} = 0.5$.



(b) Stress evolution for $\dot{\gamma} = 0.5$.

Figure 4: MATLAB post-processing results for shear rate 0.5 s^{-1}.

5.1 Comparative Interpretation

The results gives the following interpretations:

- Increasing the imposed shear rate from 0.1 to 0.5 s^{-1} scales the bulk shear stress and modifies the local velocity profile.
- The pressure profile and pressure gradient are sensitive to shear rate.
- The effective friction coefficient $\mu = S_{xy}/S_{yy}$ tracks the transition to a quasi-steady shearing regime.

6 Conclusion

This project demonstrates a structured numerical workflow for granular shear simulation. The pouring stage establishes an initial particle packing, the compression stage enforces a dense assembly at a target normal pressure, and the shear stage explores rate-dependent response. The MATLAB script extracts continuum fields from discrete particle data and makes the rate dependence of stress and flow fields explicit.

The results contained in the repository indicate that the simulation pipeline can capture both local and bulk mechanical responses: velocity profiles, concentration gradients, pressure distributions, and the evolution of normal and shear stress components.